Sample Spaces

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1. Introduction

Probability theory begins with a single primitive: a set whose elements are the possible outcomes of an experiment. This set is the *sample space*. Every subsequent concept — events, random variables, expectations, conditional distributions — is built on top of this primitive. Before we can assign probabilities, we must first agree on what we are assigning probabilities *to*.

2. Definitions

Definition 1 (Sample Space)**.** A *sample space* is a non-empty set Ω whose elements represent the mutually exclusive and collectively exhaustive outcomes of an experiment or random phenomenon.

Definition 2 (Outcome)**.** An *outcome* is an element $\omega \in \Omega$. It represents one complete atomic realization of the experiment.

Definition 3 (Event)**.** An *event* is a subset $A \subseteq \Omega$. We say the event A *occurs* on outcome ω if $\omega \in A$. In the finite case, the family of all events is the power set

$$2^{\Omega} = \{A : A \subseteq \Omega\}.$$

Example 4 (Coin flip)**.** $\Omega = \{H, T\}$. The events are $\emptyset, \{H\}, \{T\}, \{H, T\}$.

Example 5 (Die roll)**.** $\Omega = \{1, 2, 3, 4, 5, 6\}$. The event "even" is $\{2, 4, 6\}$.

Example 6 (Two dice)**.** $\Omega = \{1, \ldots, 6\}^2$, with $|\Omega| = 36$. The event "sum is 7" is $\{(1,6),(2,5),(3,4),(4,3),(5,2),(6,$

Example 7 (Infinite coin sequences)**.** $\Omega = \{H, T\}^{\mathbb{N}}$, an uncountable set. This is the natural sample space for asymptotic statements such as the strong law of large numbers.

3. The Power Set Theorem

Theorem 8 (Power Set Cardinality)**.** *If Ω is a finite set with $|\Omega| = n$, then the power set 2^{Ω} has cardinality 2^n.*

Proof. We proceed by induction on n.

Base case ($n = 0$). The unique set of cardinality 0 is $\Omega = \emptyset$. Its only subset is $\emptyset$ itself, so $|2^{\emptyset}| = 1 = 2^0$.

Inductive step. Assume the claim holds for all sets of cardinality n. Let Ω have cardinality $n+1$. Fix any element $x \in \Omega$ and let $\Omega' = \Omega \setminus \{x\}$, which has cardinality n.

Partition the subsets of Ω into two classes:

1. Subsets that do *not* contain x — these are exactly the subsets of Ω', and by the inductive hypothesis there are 2^n of them.

2. Subsets that *do* contain x — each such subset has the form $A \cup \{x\}$ where $A \subseteq \Omega'$. The map $A \mapsto A \cup \{x\}$ is a bijection between $2^{\Omega'}$ and $\{B \subseteq \Omega : x \in B\}$, so there are again 2^n such subsets.

The two classes are disjoint and exhaust 2^Ω, so

$$|2^\Omega| \;=\; 2^n + 2^n \;=\; 2 \cdot 2^n \;=\; 2^{n+1}.$$

This completes the induction. $\square$

Corollary 9. *A coin flip ($n = 2$) admits $2^2 = 4$ events; a die ($n = 6$) admits $2^6 = 64$ events; two dice ($n = 36$) admit $2^{36} \approx 6.87 \times 10^{10}$ events.*

4. Remarks on the Infinite Case

When Ω is uncountable (e.g. $\Omega = [0, 1]$), one cannot in general use the full power set 2^Ω as the event family — there exist subsets to which no translation-invariant probability measure can be consistently assigned (the Vitali construction). Instead one restricts to a σ-algebra $\mathcal{F} \subseteq 2^\Omega$, typically the Borel σ-algebra. The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is then called a *probability space*; sample spaces are the first slot of this triple. See node 18-sigma-algebras for the continuation.